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Deep Learning and Applications

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Diffusion Models

- Training objective is derived from a variational ELBO, just like VAEs
- Goal: Maximum likelihood

$$p_{\theta}(x_0) = \int_{x_T \rightarrow x_{T-1} \rightarrow \dots \rightarrow x_0} p_{\theta}(x_{0:T}) dx_{1:T} = \int p(x_T) \prod_{t=1}^T p_{\theta}(x_{t-1} | x_t) dx_{1:T} \quad \text{intractable}$$

Introduce a fixed variational distribution $q(x_{1:T} | x_0)$

$$p_{\theta}(x_0) = \int p_{\theta}(x_{0:T}) dx_{1:T} = \int q(x_{1:T} | x_0) \frac{p_{\theta}(x_{0:T})}{q(x_{1:T} | x_0)} dx_{1:T}$$

$$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_T$$

An expectation over q

$$\log p_{\theta}(x_0) = \log \int q(x_{1:T} | x_0) \frac{p_{\theta}(x_{0:T})}{q(x_{1:T} | x_0)} dx_{1:T} = \log \mathbb{E}_q \left[\frac{p_{\theta}(x_{0:T})}{q(x_{1:T} | x_0)} \right]$$

Jensen's Inequality

$$\log p_{\theta}(x_0) \geq \mathbb{E}_{q(x_{1:T} | x_0)} \left[\log \frac{p_{\theta}(x_{0:T})}{q(x_{1:T} | x_0)} \right] \quad \text{ELBO}$$

$$\begin{cases} q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}) \\ \text{Each term is predefined} \\ q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t} x_{t-1}, (1 - \alpha_t) I) \end{cases}$$

Jensen's Inequality
for the concave function log,

$$f(\text{average}) \geq \text{average}(f)$$

$$f(\mathbb{E}_q[z]) \geq \mathbb{E}_q[f(z)]$$

$$\log \mathbb{E}_q[z] \geq \mathbb{E}_q[\log z]$$



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$$\log p_{\theta}(x_0) \geq \mathbb{E}_{q(x_{1:T}|x_0)} \left[\log \frac{p_{\theta}(x_{0:T})}{q(x_{1:T}|x_0)} \right] = \underbrace{\mathbb{E}_q [\log p_{\theta}(x_{0:T})]}_{\text{generative model}} - \underbrace{\mathbb{E}_q [\log q(x_{1:T}|x_0)]}_{\text{variational posterior (forward diffusion)}}$$

Generative model

$$\log p_{\theta}(x_{0:T}) = \log p(x_T) + \underbrace{\sum_{t=1}^T \log p_{\theta}(x_{t-1} | x_t)}_{\text{generative model}} \quad \boxed{x_T \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_0}$$

$$\underbrace{\sum_{t=1}^T \log p_{\theta}(x_{t-1} | x_t)}_{\text{generative model}} = \underbrace{\log p_{\theta}(x_0 | x_1)}_{x_0 \text{ term}} + \sum_{t=2}^T \log p_{\theta}(x_{t-1} | x_t)$$

$$\log p_{\theta}(x_{0:T}) = \log p(x_T) + \underbrace{\log p_{\theta}(x_0 | x_1) + \sum_{t=2}^T \log p_{\theta}(x_{t-1} | x_t)}_{\text{joint distribution}}$$

joint distribution

$$q(a, b, c) = q(c) q(b|c) q(a|b, c)$$

Forward diffusion

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1})$$



$$\boxed{x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \cdots \rightarrow x_T}$$

$$\log q(x_{1:T} | x_0) = \log q(x_1 | x_0) + \sum_{t=2}^T \log q(x_t | x_{t-1})$$

Factorization T→1

$$= \log q(x_T | x_0) + \sum_{t=2}^T \log q(x_{t-1} | x_t, x_0)$$



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Generative model

$$\log p_{\theta}(x_{0:T}) = \log p(x_T) + \log p_{\theta}(x_0 | x_1) + \sum_{t=2}^T \log p_{\theta}(x_{t-1} | x_t)$$

Forward diffusion

$$\log q(x_{1:T} | x_0) = \log q(x_T | x_0) + \sum_{t=2}^T \log q(x_{t-1} | x_t, x_0)$$

$$\mathcal{L} = \mathbb{E}_q [\log p_{\theta}(x_{0:T}) - \log q(x_{1:T} | x_0)]$$

$$= \underbrace{\mathbb{E}_q [\log p_{\theta}(x_0 | x_1)]}_{\substack{\text{reconstruction term} \\ x_0 \text{ term (decoder)}}} + \underbrace{\mathbb{E}_q [\log p(x_T) - \log q(x_T | x_0)]}_{\text{prior matching term}} + \sum_{t=2}^T \underbrace{\log p_{\theta}(x_{t-1} | x_t)}_{\text{Denoising matching terms (for } t = 2 \dots T)} - \sum_{t=2}^T \log q(x_{t-1} | x_t, x_0)$$

VAE

$$\mathcal{L}_{\text{ELBO}}(\theta, \phi) = \underbrace{\mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x | z)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q_{\phi}(z | x) \| p(z))}_{\text{prior matching term}}$$



Diffusion Models

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_q [\log p_\theta(x_{0:T}) - \log q(x_{1:T}|x_0)] \\ &= \underbrace{\mathbb{E}_q [\log p_\theta(x_0 | x_1)]}_{\text{reconstruction term}} + \underbrace{\mathbb{E}_q [\log p(x_T) - \log q(x_T | x_0)]}_{\text{prior matching term}} + \underbrace{\sum_{t=2}^T \log p_\theta(x_{t-1} | x_t) - \sum_{t=2}^T \log q(x_{t-1} | x_t, x_0)}_{\text{Denoising matching terms (for } t = 2 \dots T)}\end{aligned}$$

- Reconstruction term
 1. In reverse diffusion, the model must recover the clean image x_0 from a noisy observation x_1 .
 2. Reverse process is probabilistic, not deterministic.

$$p_\theta(x_0 | x_1) = \mathcal{N}(\mu_\theta(x_1), \sigma^2 I)$$

3. The model predicts the **mean** of this distribution: $x_0 | x_1 \sim \mathcal{N}(\mu_\theta(x_1), \sigma^2 I)$ $\hat{x} = \mathbb{E}_{p_\theta}[x_0 | x_1]$
4. Maximizing the likelihood of this Gaussian is equivalent to minimizing mean square error.

L2 loss (MSE) $\|x_0 - \hat{x}\|_2^2$

- Prior matching term (No trainable parameters)

$$\mathbb{E}_q [\log p(x_T) - \log q(x_T | x_0)] = -D_{\text{KL}}(q(x_T | x_0) \| p(x_T)) \quad \text{model's prior} \quad p(x_T) = \mathcal{N}(0, I) \quad \nabla_\theta D_{\text{KL}} = 0$$

match the prior used by the generative model.



Diffusion Models

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_q [\log p_\theta(x_{0:T}) - \log q(x_{1:T}|x_0)] \\ &= \underbrace{\mathbb{E}_q [\log p_\theta(x_0 | x_1)]}_{\text{reconstruction term}} + \underbrace{\mathbb{E}_q [\log p(x_T) - \log q(x_T | x_0)]}_{\text{prior matching term}} + \underbrace{\sum_{t=2}^T \log p_\theta(x_{t-1} | x_t) - \sum_{t=2}^T \log q(x_{t-1} | x_t, x_0)}_{\text{Denoising matching terms (for } t = 2 \dots T)}\end{aligned}$$

- Denoising matching terms (for $t = 2 \dots T$)

$$\sum_{t=2}^T \mathbb{E}_q [\log p_\theta(x_{t-1} | x_t) - \log q(x_t | x_{t-1}, x_0)]$$

Model's reverse step Forward diffusion's true posterior

Extracting the t -th Term

$$\mathbb{E}_q [\log p_\theta(x_{t-1} | x_t) - \log q(x_{t-1} | x_t, x_0)] = -D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t))$$

This KL forces the model's reverse transition to imitate the forward diffusion's true posterior

$$\sum_{t=2}^T D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t))$$

At $t = 1$?



1. The model learns how to slightly denoise at each timestep.
2. This learning objective naturally becomes minimizing a Gaussian KL.



Diffusion Models

- Denoising matching terms

$$\mathcal{L}_{\text{denoise}} = - \sum_{t=2}^T \mathbb{E}_{q(x_t|x_0)} \left[D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_{\theta}(x_{t-1} | x_t)) \right]$$

recover x_{t-1} (a less noisy sample) from x_t (a noisier sample), and the correct target for this denoising step is the forward diffusion posterior, q

1. KL between two Gaussians in d dimensions:

$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(\mu_q(x_t, x_0), \Sigma_q(t)),$$

$$p_{\theta}(x_{t-1} | x_t) = \mathcal{N}(\mu_{\theta}(x_t, t), \Sigma_{\theta}(t)).$$

$$D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_{\theta}(x_{t-1} | x_t)) = \frac{1}{2} \left[\log \frac{\det \Sigma_{\theta}(t)}{\det \Sigma_q(t)} - d + \text{tr}(\Sigma_{\theta}(t)^{-1} \Sigma_q(t)) + (\mu_{\theta}(x_t, t) - \mu_q(x_t, x_0))^{\top} \Sigma_{\theta}(t)^{-1} (\mu_{\theta}(x_t, t) - \mu_q(x_t, x_0)) \right]$$

$$D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_{\theta}(x_{t-1} | x_t)) = \frac{1}{2\sigma^2} \|\mu_{\theta}(x_t, t) - \mu_q(x_t, x_0)\|_2^2.$$

2. Fix the covariance (practical choice)

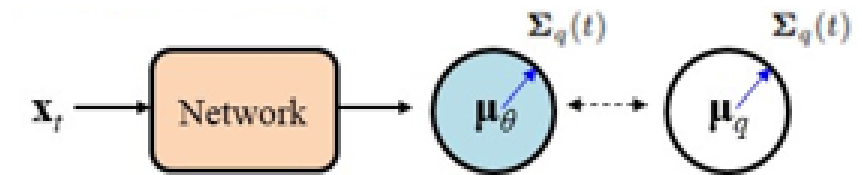
$$\Sigma_{\theta}(t) = \Sigma_q(t) = \sigma_t^2 I \quad (\text{known scalar schedule})$$

$$\log \frac{\det \Sigma_{\theta}}{\det \Sigma_q} = 0$$

$$\text{tr}(\Sigma_{\theta}^{-1} \Sigma_q) = d$$

$$\mathcal{L}_{\text{denoise}} = - \sum_{t=2}^T \mathbb{E}_{q(x_t|x_0)} \left[\frac{1}{2\sigma_t^2} \|\mu_{\theta}(x_t, t) - \mu_q(x_t, x_0)\|_2^2 \right] \quad \text{L2 loss (MSE)}$$

An MSE loss between the model mean μ_{θ} and the true posterior mean μ_q at each timestep.



Hard to Train

- Model must learn **both the mean (μ) and the covariance (Σ)**
- Training is **highly sensitive** to Σ and often **unstable**
- This difficulty occurs at **every single diffusion time step**

Diffusion Models

- Denoising matching terms

$$D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p_{\theta}(x_{t-1} | x_t)) = \frac{1}{2\sigma_t^2} \|\mu_{\theta}(x_t, t) - \mu_q(x_t, x_0)\|_2^2.$$

- Forward diffusion formula

the foundation of all diffusion derivations

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad x_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon}{\sqrt{\bar{\alpha}_t}}$$

Forward posterior mean

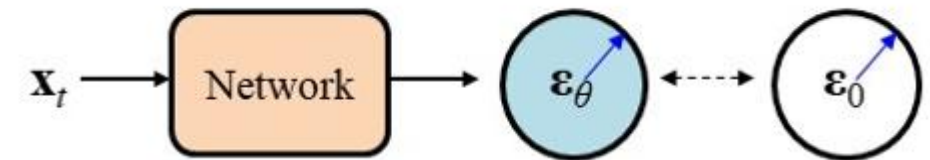
$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(\mu_q(x_t, x_0), \Sigma_q(t))$$

$$\mu_q(x_t, x_0) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right) \quad \text{the true posterior mean written as a function of } \epsilon$$

- Reverse model mean

$$p_{\theta}(x_{t-1} | x_t) = \mathcal{N}(\mu_{\theta}(x_t, t), \Sigma_t)$$

$$\mu_{\theta}(x_t, t) = \frac{1}{\sqrt{\alpha_t}} (x_t - c_t \epsilon_{\theta}(x_t, t)) \quad \begin{aligned} \epsilon_{\theta}(x_t, t) &= \text{predicted noise} \\ c_t &= \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \end{aligned}$$



μ_{θ} and μ_q have the same functional form —the only difference is ϵ_{θ} versus true ϵ .

$$\mu_{\theta} = A(x_t) - B \epsilon_{\theta}, \quad \mu_q = A(x_t) - B \epsilon,$$

$$\|\mu_{\theta} - \mu_q\|^2 = C_t \|\epsilon_{\theta}(x_t, t) - \epsilon\|^2$$

KL divergence (a μ -difference) naturally becomes an ϵ -difference (MSE).

$$\mathcal{L} = \mathbb{E}_{t, x_0, \epsilon} [\|\epsilon - \epsilon_{\theta}(x_t, t)\|^2]$$

Diffusion Models

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_q [\log p_\theta(x_{0:T}) - \log q(x_{1:T}|x_0)] \\ &= \underbrace{\mathbb{E}_q [\log p_\theta(x_0 | x_1)]}_{\text{reconstruction term (fixed variance } \alpha_1 \approx 1)}} + \underbrace{\mathbb{E}_q [\log p(x_T) - \log q(x_T | x_0)]}_{\text{prior matching term (KL = 0)}} + \sum_{t=2}^T \log p_\theta(x_{t-1} | x_t) - \sum_{t=2}^T \log q(x_{t-1} | x_t, x_0)\end{aligned}$$

Denoising matching terms (for t = 2...T)

- ϵ -prediction (Natural Denoising ---- A Simple & Stable MSE Training Objective)
 1. Likelihood-based theory
 2. Training becomes a simple and stable MSE loss
 - a. No GAN min-max
 - b. No VAE log-variance trick $\epsilon \sim \mathcal{N}(0, I)$
 3. Learning ϵ is extremely easy
 - a. GAN $\rightarrow$ must learn full image distribution
 - b. VAE $\rightarrow$ posterior collapse issues